

AERIS Operator Cheat Sheet

A) Core CLI — day-to-day

1. Check installation/version

```
aeris version
```

Use when you want to confirm the package/CLI is alive.

2. Top-level help

```
aeris --help
```

3. Geometry command help

```
aeris geometry --help
```

4. Dataset command help

```
aeris dataset --help
```

5. Pipeline command help

```
aeris pipeline --help
```

6. Generate one baseline geometry

```
aeris geometry generate -c configs/geometry/baseline_bwb.yaml
```

7. Generate one random geometry from design space

```
aeris geometry generate -c configs/geometry/wing_bwb.yaml
```

8. Small LHS dataset

```
aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 20 --  
sampler lhs_v1 --sampler-seed 123
```

9. Fast production dataset, no plots

```
aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 500 --  
sampler lhs_v1 --sampler-seed 123 --no-save-plot --name  
stress_wing_bwb_n500_lhs123_noplots
```

10. AeroSandbox-enabled batch

```
aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 20 --  
sampler lhs_v1 --sampler-seed 123 --no-save-plot --build-aerosandbox --  
name stress_wing_bwb_n20_lhs123_asb
```

11. Random sampler test

```
aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 20 --  
sampler random_v1 --sampler-seed 123 --no-save-plot --name  
random_test_n20
```

12. Inspect dataset

```
aeris dataset inspect --dataset  
data/datasets/stress_wing_bwb_n500_lhs123_noplots
```

13. Inspect AeroSandbox-enabled dataset

```
aeris dataset inspect --dataset  
data/datasets/stress_wing_bwb_n20_lhs123_asb
```

14. Smoke test, good config

```
aeris pipeline smoke -c configs/geometry/baseline_bwb.yaml
```

15. Smoke test, bad config

```
aeris pipeline smoke -c configs/smoke/bad.yaml
```

B) Useful scripts — diagnostics / visualization / checks

16. Visualize baseline airplane

```
python scripts/bwb_segmented_v1/visualize_airplane.py --mode baseline
```

17. Visualize one dataset airplane

```
python scripts/bwb_segmented_v1/visualize_airplane.py --mode dataset --  
dataset data/datasets/stress_wing_bwb_n500_lhs123_asb --geometry-id  
geom_00001
```

18. Visualize 5 random airplanes sequentially

```
DATASET="data/datasets/stress_wing_bwb_n500_lhs123_asb"  
ls "$DATASET/geometry" | shuf | head -n 5 | while read GEOM_ID; do  
  echo "Visualizing $GEOM_ID"  
  python scripts/bwb_segmented_v1/visualize_airplane.py \  
    --mode dataset \  
    --dataset "$DATASET" \  
done
```

```
--geometry-id "$GEOM_ID"  
done
```

19. Run bad-config failure check script

```
python scripts/generic/check_bad_config_failure.py
```

20. Run architecture integrity check

```
python scripts/generic/check_architecture_integrity.py
```

21. Run metadata regeneration check

```
python scripts/generic/check_metadata_regeneration.py
```

22. Run seed variation check

```
python scripts/generic/check_seed_variation.py
```

23. Run large dataset sanity check

```
python scripts/generic/check_large_dataset_sanity.py
```

24. Audit a dataset

```
python scripts/generic/audit_dataset.py
```

25. Run one case and save debug artifacts/plot

```
python scripts/bwb_segmented_v1/run_single_case_and_plot_3d.py
```

C) Extra very useful shell helpers

These are not AERIS commands, but they save time.

Latest dataset path

```
latest_dataset="$(ls -td data/datasets/* | head -n 1)"  
echo "$latest_dataset"
```

Count geometry folders in dataset

```
find "$latest_dataset/geometry" -maxdepth 1 -type d | wc -l
```

Metadata row count

```
wc -l "$latest_dataset/metadata.csv"
```

Failures row count

```
wc -l "$latest_dataset/failures.csv"
```

Tail dataset log

```
tail -n 20 "$latest_dataset/logs/dataset.log"
```

Recommended grouping for you

Use most often

These are your real everyday commands:

```
aeris version
aeris geometry generate -c configs/geometry/baseline_bwb.yaml
aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 20 --
sampler lhs_v1 --sampler-seed 123
aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 500 --
sampler lhs_v1 --sampler-seed 123 --no-save-plot
aeris dataset inspect --dataset <dataset_path>
python scripts/bwb_segmented_v1/visualize_airplane.py --mode dataset --
dataset <dataset_root> --geometry-id <geom_id>
```

Use when debugging

```
aeris pipeline smoke -c configs/geometry/baseline_bwb.yaml
python scripts/generic/check_bad_config_failure.py
python scripts/generic/check_architecture_integrity.py
python scripts/generic/check_metadata_regeneration.py
```